CS2610: Computer Organization and Architecture Lab

Lab 4: Simple multicycle RISC-V design

23rd February 2026

Objective

In this lab, you will be working with simple RISC-V core design shared in moodle. The aim of this assignment is to implement control flow instructions in the given design, and show it's working.

1 Question

The following instructions are to be implemented :

1. Conditional branches : BEQ, BNE, BLT, BGE, BLTU, BGEU
2. JAL
3. JALR

2 Ideal steps to follow

- Understand the modules within the design, especially the interface and control signals.
- Ideate what kind of changes are required for each module - ex. add decode logic for each instruction in decoder module, etc.
- Refer to Unpriveleged ISA manual/RISC-V card for exact instruction encoding for each instruction.
- Implement each instruction and test them out one by one.

3 How to test

The zip file shared has a setup to compile a c code/ ASM code and run it on the design. Test the design on different programs with loop.

4 Submission Instructions

1. The assignment must be completed individually.
2. This assignment is a take home one, viva will be held in next lab.
3. The following artifacts must be submitted:

- Screenshots of all the modifications in each module.
- Sample test code with the screenshot of this test code running.
- The updated logisim circuit file.

4. Compress all files into a single ZIP archive and submit it on Moodle.